

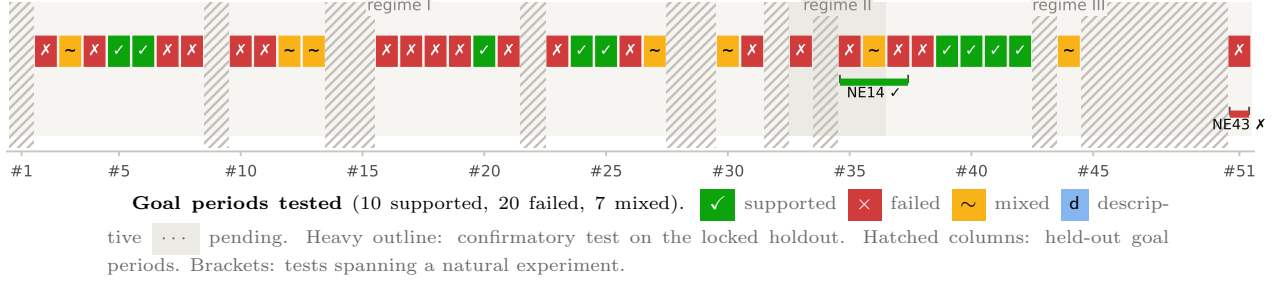
H38: Joint silences are platform stalls

Vivian Rogers, with Claude Opus 5.5 · agent-swarm physics · Updated 2026-10-04

Complete 50% Faithfulness 2/5 Usefulness 2.5/5

Bottom line. Joint silences are rare and operator-scheduled, not platform stalls; in the always-on regime, 70–80% of apparent co-activation is the runner that starts and stops all agents together.

level: hypothesis · pipeline checks 86% · updated 2026-10-04



WHAT HAPPENED

H38 builds a shared outage table: minutes with ≤ 1 active agent, with causes from logs. It asks, in 35 non-holdout periods, how much of H02/H19's collective co-activation is infrastructure. **Round 1b** re-runs everything on the corrected activity table (round 1 lost half of all events) with DQ8's calibrated null. **Joint silences are rare and scheduled:** 5.3% of minutes (was 14.3%), 78% outside the operator's run. Error bursts make them *less* likely. **In the always-on regime III, most co-activation is day-edge synchrony** ($f = 0.72$; 0.83 after trimming). Trimming each day to the common running window leaves a significant excess only in #44 and #51 (and #37, marginally). Regime I keeps the excess in 16/18 periods. In the native test NE43, the edge share does not drop when the operator's pause/resume messages stop, so the edges are the runner's schedule. The holdout is not run.

PHYSICAL MODEL

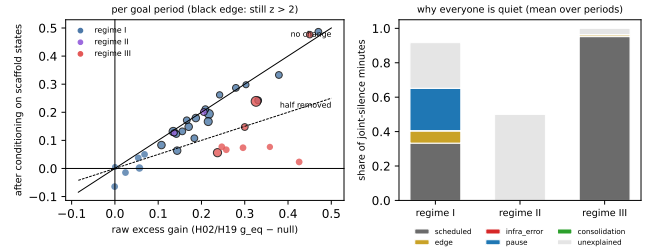
Models 01 (Curie–Weiss) and 09 (Hawkes). The spin $s_i(t) = \pm 1$ marks agent i active in minute t . $h_i(b)$ is a per-agent field per 30-min block. The stall indicator $z(t)$ is a second field that switches agents off: operator off, not yet started or finished, consolidating, or infrastructure error. π is the stall fraction and p the on-rate. The order parameter is the gain g . The stall-adjusted gain trims to the window in which all agents run and sets scaffold-unavailable agent-minutes to the block mean. Block-shift surrogates, drawn on the kept rows, give the null.

$$P(\mathbf{s}) \propto \exp \left[\frac{\beta J_0}{2N} M^2 + \sum_i (h_i(b) + \Delta_i z(t)) s_i \right], \quad M = \sum_i s_i$$

$$\text{VR} = \frac{\sum_b n_b \text{Var}_b M}{\sum_b n_b \sum_i \text{Var}_b s_i}, \quad g = 1 - \text{VR}^{-1}$$

stalls alone: $\text{VR} - 1 \approx \frac{(N-1)\pi p}{1 - (1-\pi)p}$ ($N=12, p=0.5, \pi=0.05 \Rightarrow g \approx 0.34$)

$$f = 1 - \frac{g_{\text{adj}} - \langle g_{\text{adj}} \rangle_{\text{null}}}{g_{\text{raw}} - \langle g_{\text{raw}} \rangle_{\text{null}}}$$



Observables. Corrected tables. (a) Excess equal-time gain per period before and after conditioning on scaffold states (dashed: half removed). Regime-III points (red) drop to or below the dashed line, except #44 and #51. Regime I (blue) stays near the identity line. Black edge: still $z > 2$. (b) Causes of joint-silence minutes, mean over periods. With the dropped events restored, operator-scheduled minutes dominate; infrastructure errors are about 0.1%.

SO WHAT? PRACTICAL VALUE FOR UNDERSTANDING AGENT SWARMS

An always-on swarm with a daily start and stop shows synchrony that is mostly its own runner schedule. This holds for mean-field coupling, the market-mode eigenvalue and co-activation. Agents boot and stop together whether or not the operator announces it. Before you draw surrogates, trim each day to the window in which all agents run. Here that removes about 80% of regime-III co-activation. Real excess remains only where agents co-move (#44, #51). Do not alert on joint silences as a stall signal: error logs do not predict them, and provider outages leave no trace. Message-driven swarms (regime I) gain nothing from this rule.

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RESULTS VS PREDICTIONS

Prediction	Round 1 → round 1b (corrected tables)	Verdict
P1 JS share 0.05–0.30, median 0.10 → 0.009; excess 0.008 above independence	→ 0.007	×
P2a scaffold reasons in JS above chance	explained 0.51 → 0.88 (mostly scheduled); > surrogate in 23 → 24/35	×
P3 infra bursts precede JS	log OR $-0.43 \rightarrow -0.96$; significant > 0 in 0 → 1/30	×
P4 stall filter removes $\geq$ half	f 0.25 → 0.26 (III 0.63 → 0.56)	×
Amend. 2 scaffold conditioning	f_{III} 0.68 → 0.72; f_1 0.11 → 0.11	III ✓
DQ8 null (trim first)	significant 29 raw → 20/35 new trimmed; III 3/8, I 16/18	
P5 λ_1 drop tracks stall share	ρ 0.89 → 0.87; ratio 0.48 → 1.00; $\sim \rightarrow$ ✓ trimmed edge: 19/41 units	✓
P6 III–I contrast shrinks $\geq$ half	+0.125 → +0.017; now +0.155 → -0.006	✓
P9 NE14 jump carried by stalls	+0.15 → -0.06; now +0.12 [-0.04, 0.27] → -0.14	✓
NE43 bookends stop (native)	edge share 0.19 → 0.14 (p 0.08); start spread narrower	×

GOAL PERIODS IN DETAIL

G02 × 1b: failed (round 1: failed; corrected tables, same r...
G03 ~ 1b: mixed (round 1: supported; corrected tables, same...
G04 × 1b: failed (replication, round 1: mixed; corrected ta...
G05 ✓ 1b: supported (round 1: failed; corrected tables, sam...
G06 ✓ 1b: supported (round 1: supported; corrected tables, ...
G07 × 1b: failed (round 1: failed; corrected tables, same r...
G08 × 1b: failed (round 1: mixed; corrected tables, same ru...
G10 × 1b: failed (round 1: failed; corrected tables, same r...
G11 × 1b: failed (round 1: failed; corrected tables, same r...
G12 ~ 1b: mixed (round 1: failed; corrected tables, same ru...
+27 more periods: see the card.

FAITHFULNESS SCORECARD

A mapping	1	F identifiability	1
B assumptions	1	G ground truth	1
C adequacy	1	H comparative	1
D unfitted predictions	1	I transfer	1
E interventional	1		

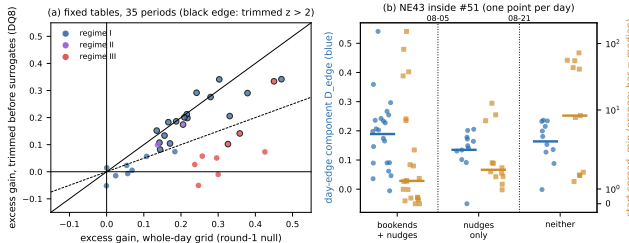
Level: **hypothesis**. Scale 0 not done or failed, 1 partial, 2 passed (`writeup/paper.tex`).

CAVEATS

LLM API failures leave no log line, so provider outages remain untested. On the corrected tables most periods have only a few joint-silence minutes. The per-period “explained above chance” test and 12 per-period verdict changes (mostly regime I) rest on these few minutes. The headline estimator (scaffold conditioning) came after synthetic smoke runs; the pre-registered stall filter fails. The **scheduled** flag comes from operator messages and is empty after 08-05; agent-level trimming does not depend on it. A whole-block stall (the 300-min stop on 2025-06-18) leaves no trace in the block-detrended gain. The holdout is not run.

ROUND 2

Second figure. (a) Excess gain on the whole-day grid (round-1 null) and after trimming to the all-present window before surrogates are drawn (DQ8). Regime III collapses toward zero; regime I stays. (b) NE43 inside #51, one point per day. The day-edge component (blue) does not fall when the bookend messages stop on 08-05. The start spread (orange) widens only when the nudger stops on 08-21 (post hoc).



Run a per-provider silence test for API outages: all agents of one lab silent while the others act. Add the trimmed (DQ8) variants to `confirm.py` before it runs. Study the #44 and #51 residual coupling. Replace the message-based **scheduled** flag with the runner’s start and stop times.

No round-2 redirects yet.

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fra/summaries/build-summaries.py · card hypotheses/H38-platform-
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